INCREMENTALITY STUDY

Did the Ads Actually Work?

What retargeting caused in a randomised test of 13,979,592 users, and what to do about it

Sharath Sista · October 2026

Independent analysis of public data (Criteo Uplift v2.1, CC BY-NC-SA 4.0; Iowa Liquor Sales). Not affiliated with or endorsed by Criteo, Google or the State of Iowa.

THE ANSWER FIRST

Three decisions this test can inform

SHOULD WE TRUST ATTRIBUTION?

Not on its own

An estimated 54 of every 100 view-through-credited conversions were caused by the ads (50 to 58). Attributed ROAS overstates incremental ROAS by about 1.9x.

SHOULD WE MOVE BUDGET?

Probably, but test first

In the scenario: dashboard ROAS 8, margin 30%, the weaker half of spend makes -£0.58 per £1. That comes from a model, so it is a hypothesis.

WHICH EXPERIMENT NEXT?

A holdout in the weaker half

A 50/50 test inside that segment settles it with about 0.4 million users. Then keep an always-on holdout.

How robust? 7 adjustment methods give 51 to 58 caused per 100, about as wide as the statistical interval

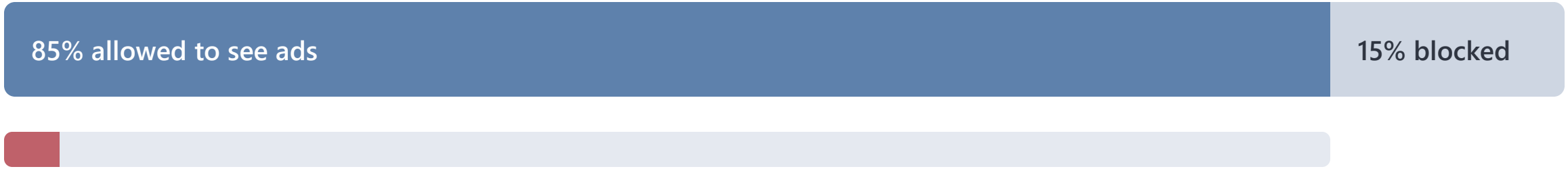
Money is a scenario. The data has no prices; ROAS and margin are assumed and varied, and the break-even rule needs neither

Benchmark levels. Criteo sub-sampled the file; the methods carry over, the numbers must be measured again

Source: Criteo Uplift v2.1 (Diemert et al.), CC BY-NC-SA 4.0 · levels describe the sub-sampled benchmark

THE EVIDENCE

A randomised test on 13,979,592 users, checked before trusting it



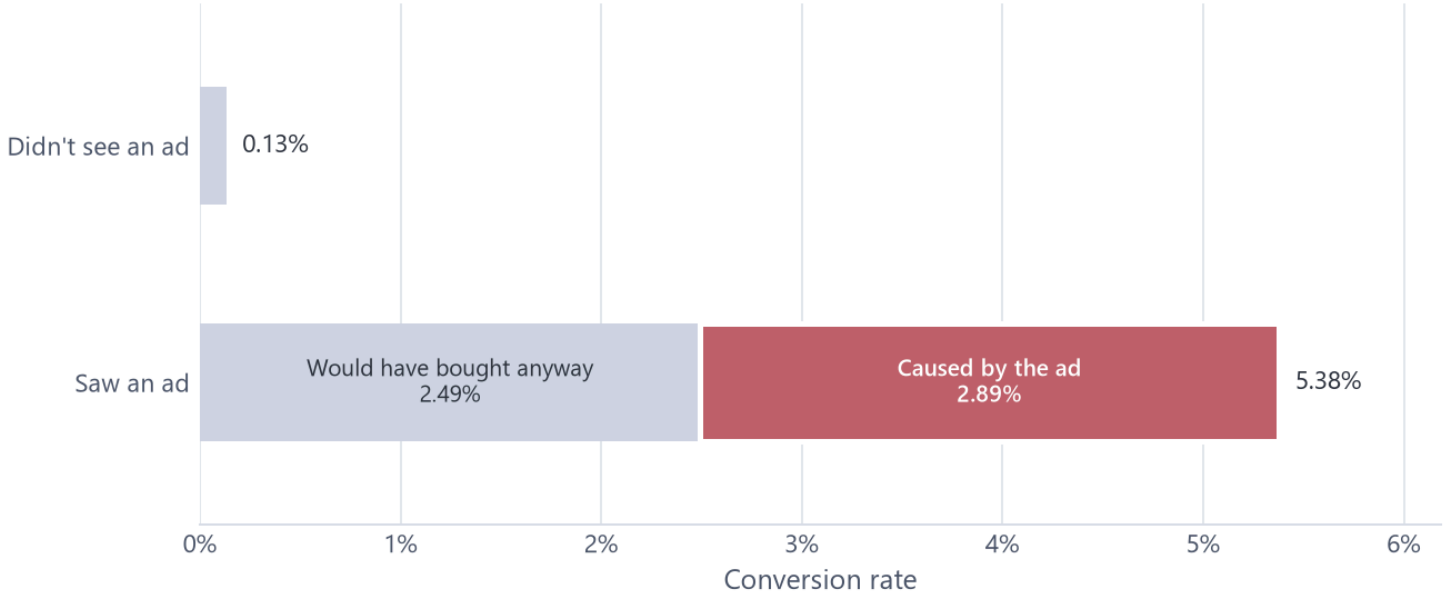
Only 3.6% of the allowed group actually saw an ad (red). Because the split is random, the difference between the groups estimates what the ads caused.

- A small leak from merged advertiser tests (classifier AUC 0.509, placebo 0.499), corrected by adjusting for features logged before any ad
- 7 adjustment methods put conversion lift at +50% to +56%; the raw difference reads +59%
- Criteo sub-sampled the data, so levels describe this benchmark; the methods transfer

Source: Criteo Uplift v2.1 (Diemert et al.), CC BY-NC-SA 4.0 · levels describe the sub-sampled benchmark

Seeing an ad looks like a 41x boost. The estimated causal effect is 2.2x

Users who saw an ad were already 19x more likely to buy. The test separates that from the ad.



Source: Criteo Uplift v2.1, 13,979,592 users. Rates describe this sub-sampled benchmark file, not Criteo's live results.

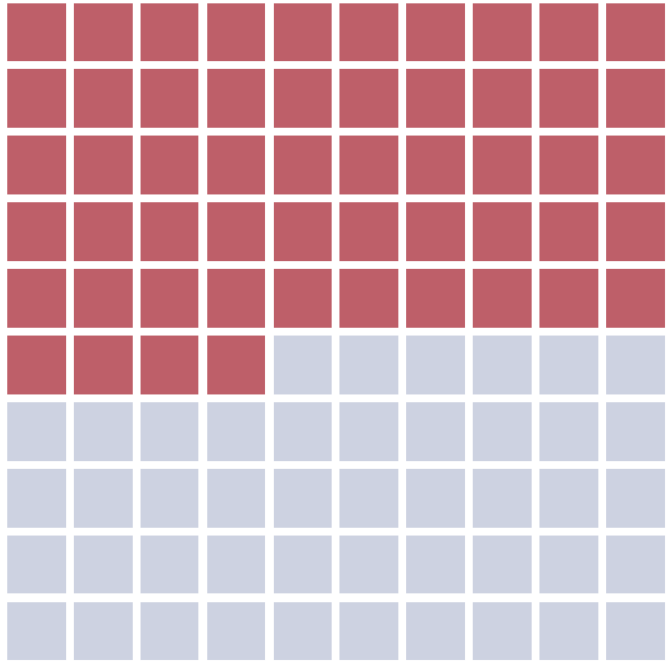
2.2x

estimated causal effect of seeing an ad (CACE)

- Saw an ad vs didn't looks like 41x
- Those users were already 19x likelier to buy
- Seeing an ad: 2.49% to 5.38% (+2.89 points)

An estimated 54 of every 100 credited conversions were caused by the ads

View-through attribution credits every conversion after an ad view.
Split by the randomised test (doubly robust, under the test's assumptions).



54 caused by the ad

46 would have happened without the ad

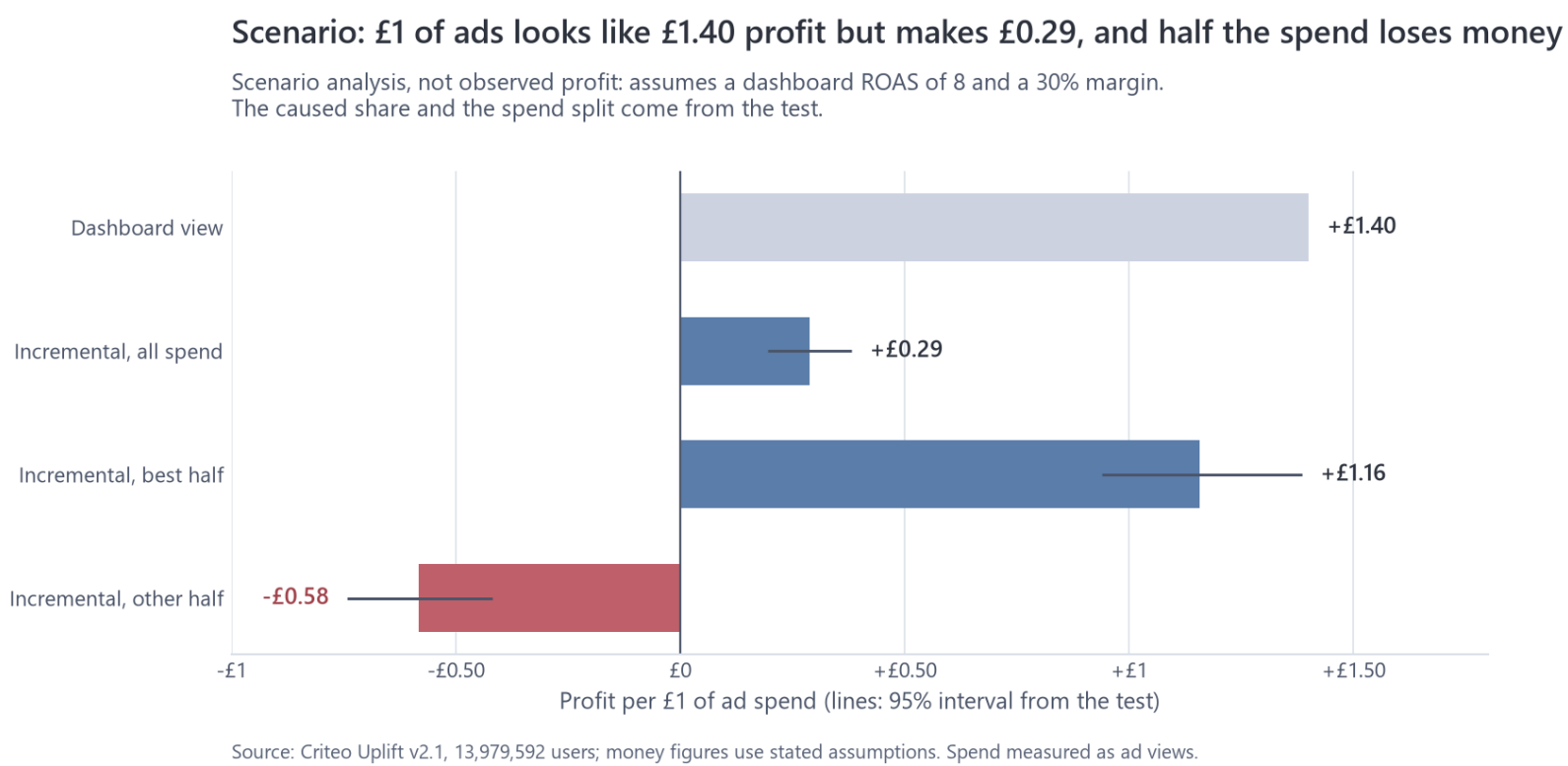
95% interval: 50 to 58 caused.
Attribution overstates the ads' conversions by about 1.9x.
Without correcting the leak from finding 1, it would read 59.

Source: Criteo Uplift v2.1, 13,979,592 users. Shares describe this sub-sampled benchmark file.

54

of every 100 view-through-credited conversions were caused by the ads (estimated)

- 95% interval 50 to 58; across the adjustment methods, 51 to 58
- Incremental ROAS = dashboard ROAS × 0.54: about 1.9x lower
- Assumes the leak runs through the recorded features, and that ads are the only difference between groups



Source: Criteo Uplift v2.1, 13,979,592 users; money figures use stated assumptions. Spend measured as ad views.

- Top 20% of users hold 91% of the effect; half the spend, aimed at likely buyers, keeps 84%
- £1 looks like +£1.40 but makes an incremental +£0.29; the weaker half makes -£0.58
- A hypothesis: auction prices, ad frequency and audience mix can shift once targeting changes

DECISION 3 · WHICH EXPERIMENT NEXT?

Test the cut first, then keep an always-on holdout

1	**A 50/50 holdout inside the weaker half**	89% of users, half the ad views. Predicted lift +55%; cut if the upper bound is below the +131% break-even. About 0.4 million users
2	**An always-on 10% holdout**	4.9 million users to see a +10% conversion lift; visits read early with 0.9 million
3	**Report incremental ROAS next to attributed ROAS**	Break-even dashboard ROAS 6.2 at a 30% margin
4	**Geo tests only where holdouts are impossible**	Methods benchmark: even 20 counties for 12 weeks detect only about 9%

And a marketing mix model for cross-channel budgets, anchored by these experiments

Code, notebooks, dashboard and report

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Sharath Sista · Data Analyst

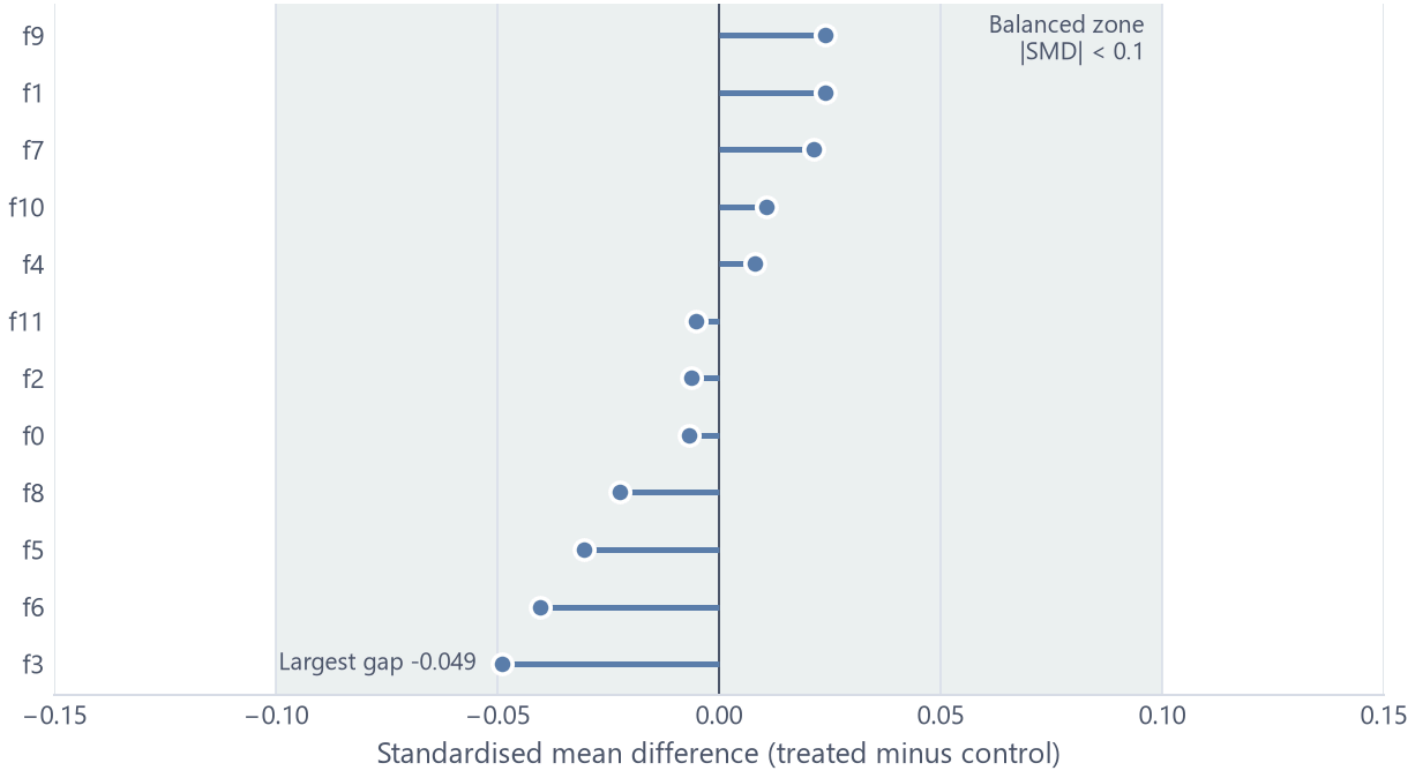
APPENDIX

The detail behind the decisions

Experiment checks · sensitivity · break-even rule · targeting · geo methods benchmark · holdout sizing · methods and limits

Treated and control users look alike on all 12 features

Gap between group means in standard deviations, 13,979,592 users. Every gap is under half the 0.1 line.



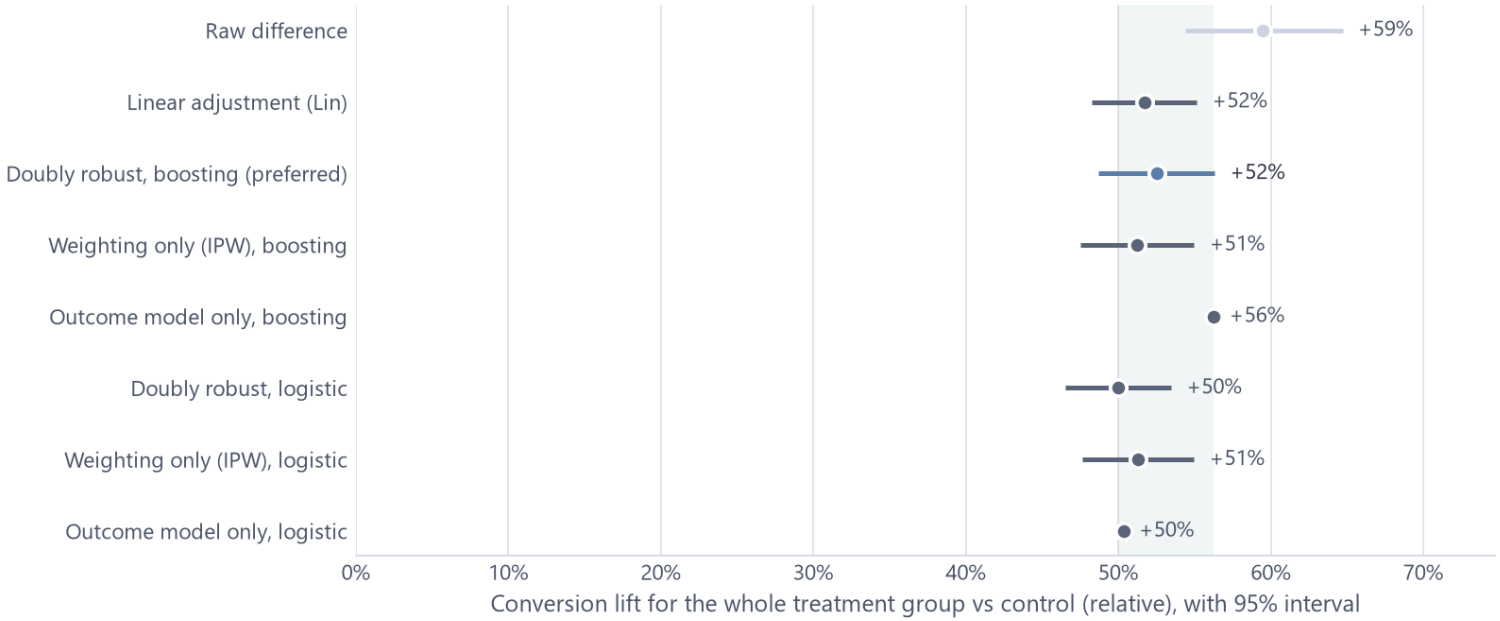
Source: Criteo Uplift v2.1 (Diemert et al., AdKDD 2018), randomised test on 13,979,592 users.

- Split exactly 85.00% treated, as designed
- No feature differs by more than 0.049 standard deviations
- Yet a classifier tells the groups apart slightly: AUC 0.509 (placebo 0.499)
- Cause: several advertisers' tests merged with different treated shares

Source: Criteo Uplift v2.1 (Diemert et al.), CC BY-NC-SA 4.0 · levels describe the sub-sampled benchmark

Any of 7 adjustment methods puts conversion lift at +50% to +56%

The raw difference reads +59%. Weighting by the fitted chance of treatment shrinks the largest feature gap from 0.049 to 0.002 standard deviations. Outcome-only models have no interval here.



Source: Criteo Uplift v2.1 (Diemert et al., AdKDD 2018), randomised test on 13,979,592 users.

- 7 adjusted specifications: conversion lift +50% to +56%, visits +19% to +20%
- Caused per 100 credited: 51 to 58
- Weighting shrinks the largest feature gap to 0.002 standard deviations
- Untestable: a leak through something the file doesn't record

A3 · A BREAK-EVEN RULE ANYONE CAN USE

At a 30% margin, retargeting needs a dashboard ROAS above 6.2 to pay

6.2
Break-even ROAS, all spend

3.7
Best half of spend

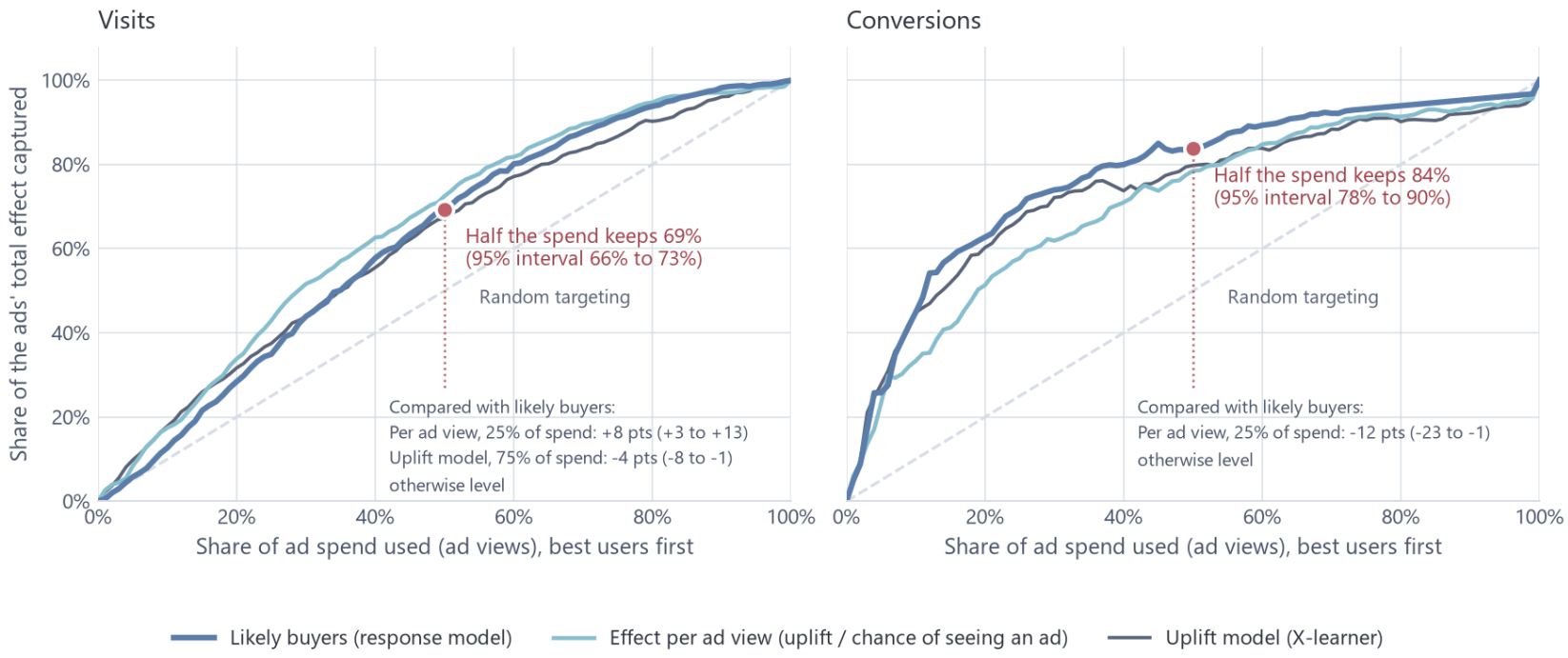
19.2
Weaker half of spend

Incremental ROAS = dashboard ROAS × **0.54**, the caused share from a holdout. Break-even needs no ROAS assumption.

Source: Criteo Uplift v2.1 (Diemert et al.), CC BY-NC-SA 4.0 · levels describe the sub-sampled benchmark

Half the ad spend, aimed at likely buyers, keeps 84% of the conversions ads cause

And 69% of the visits. Spend measured as ad views. Effect-per-ad-view ranking is exploratory (added after first results).



Source: Criteo Uplift v2.1, 13,979,592 users (80% train, 20% test). Shares describe this sub-sampled benchmark file.

- The top 20% of users hold 91% of the effect
- Half the spend, aimed at likely buyers, keeps 84% of conversions caused (78% to 90%)
- A simple likely-buyer model matched T- and X-learner uplift models on conversions

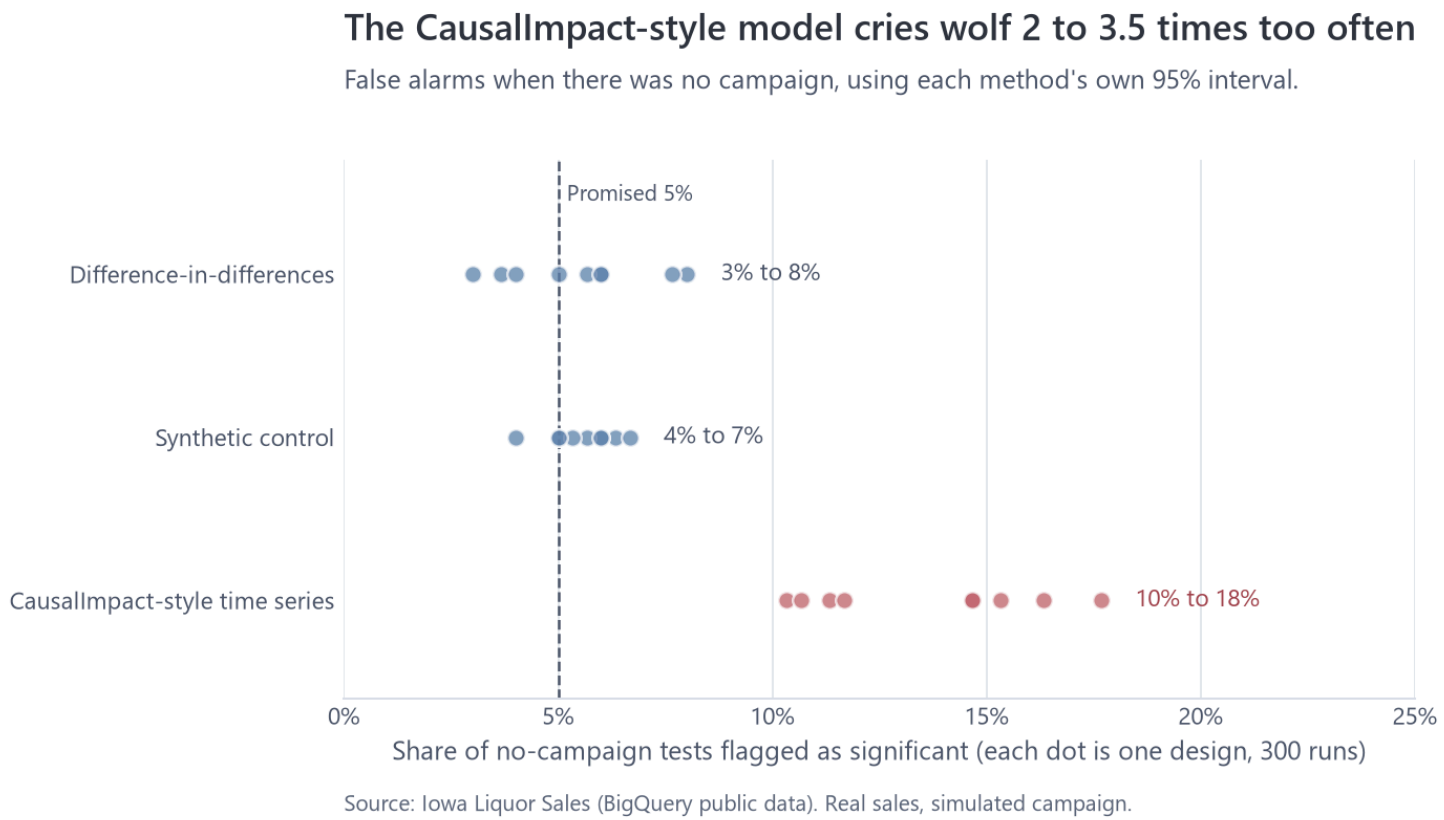
Even 20 counties for 12 weeks only reliably detect a lift of about 9%

Methods demonstration: real Iowa liquor sales, simulated campaigns. 300 runs per design, all methods held to a 5% false-alarm rate.

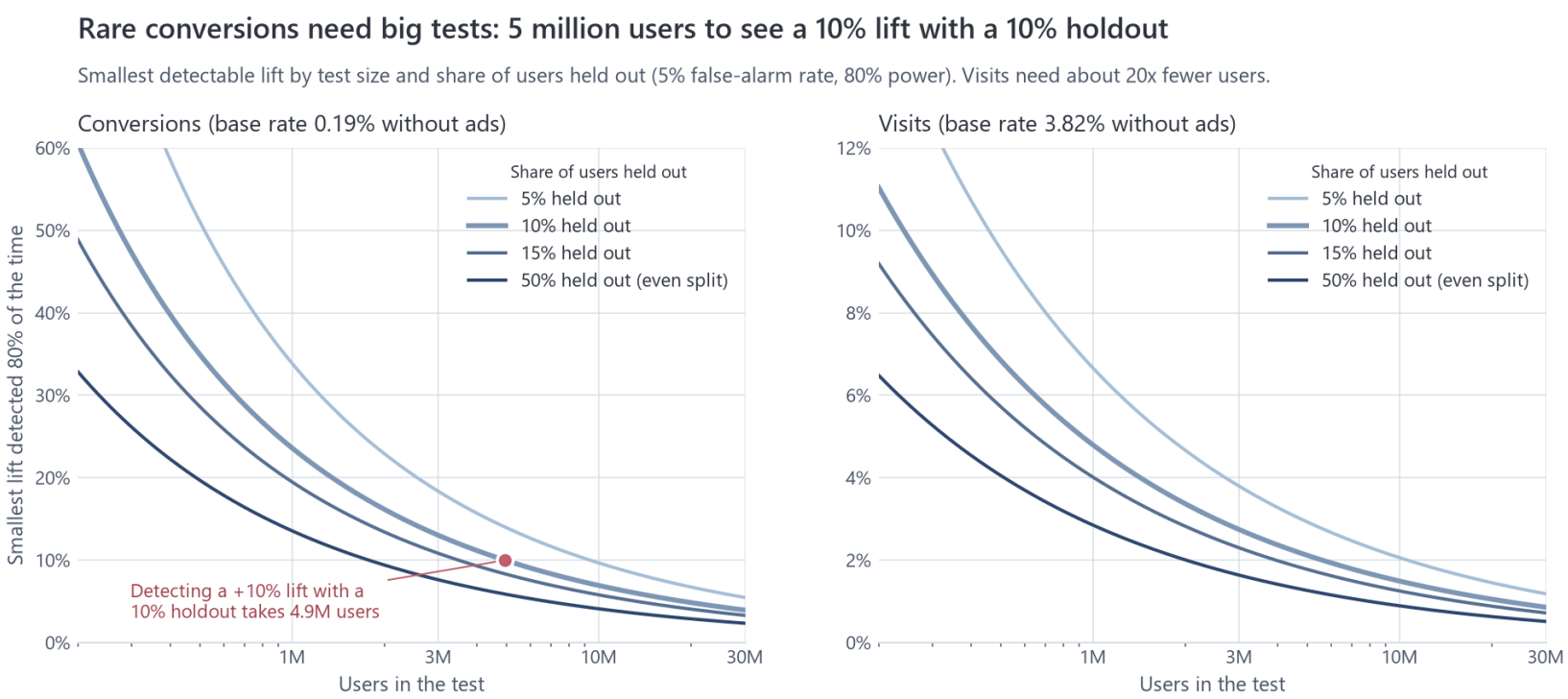


Source: Iowa Liquor Sales (BigQuery public data). Real sales, simulated campaign.

- 2,700 simulated campaigns on real Iowa county sales, so the true answer is known
- Best design (20 counties, 12 weeks) reliably detects about 9%
- Assumes a uniform lift, random counties and no spillover: a best case
- More regions help faster than more weeks



- Difference-in-differences: 3% to 8% false alarms
- Synthetic control: 4% to 7%
- CausalImpact-style: 10% to 18%, promised 5%
- Placebo-test any model before trusting its intervals



Base rates: Criteo Uplift v2.1 control group. Formula checked against the Criteo test's own precision (within 0.2 points).

- About 4.9 million users to see a +10% conversion lift with a 10% holdout
- 1.8 million with a 50/50 split
- Visits read early with about 0.9 million

What this does and doesn't show

- Methods: intent-to-treat; Lin and doubly robust (AIPW, cross-fitted) adjustment; two-stage least squares (CACE); T- and X-learner uplift on 2.8M held-out users; DiD, synthetic control, structural time series
- Criteo sub-sampled the file: levels describe this benchmark; the methods transfer
- No prices or costs: money figures are scenarios, with sensitivity tables and a break-even rule
- Spend is proxied by ad views; real cost depends on frequency and auction prices
- Geo results are a methods benchmark on one lumpy dataset, not a measured campaign